

Tue, Mar 3 at 12:16

Lance Kuo-Esser

I am getting increasingly convinced this is worth building.

The goal would be to develop a real MVP and run 1 to 2 pilots within the next year. The first phase would focus on curating a high-quality dataset for 100 well-characterized compounds from public regulatory and literature sources using a semi-manual process with consistent units and clear labeling rules. From there, we would move into model development for dominant enzyme prediction and DDI risk ranking, followed by a simple UI that outputs trial-relevant flags and recommended validation experiments. The target would be to identify pilot partners by late fall.

This will require meaningful time and follow-through, so I only want to involve people who are genuinely interested and able to contribute consistently. If you would like to be part of it and can commit the time and energy, let me know. If not, no worries we can meet on the next project

LK

Tue, Mar 3 at 13:30

Yea sounds good

Tue, Mar 3 at 14:30

Shayne Skrtic

I'm very down I've actually already got some good training code that could be used for dominant enzyme prediction. If you put together/have a list of parameters for the dataset you think will be best to train on I can start getting some scraping started early next week

SS

Tue, Mar 10 at 14:08

Lance Kuo-Esser

send emails

LK

bvomartin@gmail.com

Shayne Skrtic

SS

Skrticshayne@gmail.com

Lance Kuo-Esser

LK

happy hunting and please watch the fucking units

LK

i'm sure questions will come up so slap em here and we can figure best way forward

LK

lmk if you guys figure a good way to scrape

Tue, Mar 10 at 22:03

Shayne Skrtic

SS

When multiple pH experimental LogP values want to standardize at 7? Or 9?

Lance Kuo-Esser

i have no clue, is there a good reason to do 9, 7 is closer to physiology but wtf do i know

LK

martin?

7

I'm not a doctor tho

Lance Kuo-Esser

no fr we know that



LK

Shayne Skrtic

SS

Lmao okay 7

Lance Kuo-Esser

LK

are you scraping this shit or just going ham rn

Shayne Skrtic

SS

Scraping the easy ones but going one at a time for some of the ones that I don't feel like thinking about a good scraper for rn lol

Lance Kuo-Esser

LK

well if it's easy enough toss ours into the easy ones to scrape

Mon, Mar 16 at 11:30

Shayne Skrtic

SS

There is a lot of variance in experimental data for LogP especially on pH, if we end up training on that im worried the noise will make the feature pretty useless. What do we think about normalizing to the XLogP3 calc for consistency instead of accuracy?

Lance Kuo-Esser

LK

what do they do for that calculation

[1 Reply](#)

Mon, Mar 16 at 16:36

Lance Kuo-Esser

LK

and martin, what info did you need for validation of binding

LK

what do they do for that calculation

Shayne Skrtic

SS

Nvm you right it's actually a silly calculation it starts at a similar structure and makes linear moves from there so not much better in terms of consistency experimental probably better

I would like a set of data that we can use to see if our code produces the same results

code produces the same results

We need something to double check our work

Lance Kuo-Esser

LK

yes so like i hat value and units do you need

Yes

Lance Kuo-Esser

LK

what

Did you ask if I need the values?

The units and values that I need are what the results should be. So for the curating part, I would need an example of what the user would input in the application and then what the results it should output so I can have a way to double check the program

Mon, Mar 16 at 19:13

Lance Kuo-Esser

LK

i guess a compound and how well it binds to a given metabolizing enzyme

Tue, Mar 17 at 06:55

So kinda like what I was doing a couple months ago?

With ketoconazole and cyp3a4

Lance Kuo-Esser

LK

ball park how may examples you think there are to find

Tue, Mar 17 at 15:20

Just need one solid pairing that is backed up by real data and from valid sources

Lance Kuo-Esser

LK

ok so we need a drug and a Ki or a delta G?

Which ever one you want it to give

Lance Kuo-Esser

LK

ok let's hold off on data adding till martin finds out where he gets to take it in the ass

Shayne Skrtic

SS

What part is this for I'm lost? If you're looking for a dataset of binding coefficients I had a side project a few years ago where I was modeling this built of Rosetta and I have a dataset I put together for this somewhere

Lance Kuo-Esser

LK

need compounds SIMLES, enzyme, and binding affinity

LK

basically can we put a smiles value in and get out binding affinity for a enzyme

Shayne Skrtic

Oh I have code for that not smiles but that's an easy update. It's not an mlm it's an alg so it takes like 3 ish minutes to run but we can preprocess.

skrticshayne/
PyRosettaDDGfolding



1

Contributor

0

Issues

0

Stars

0

Forks



SS

skrticshayne/PyRosettaDDGfolding
github.com

Lance Kuo-Esser

LK

what do you need

Shayne Skrtic

Shayne Skrtic

That does folding too so I'd just axe that part if you wanted to go with that

SS

Just compound and enzyme name

Lance Kuo-Esser

LK

did you validate its accuracy

Tue, Mar 17 at 18:25

Shayne Skrtic

SS

Once upon a time yeah, I'll find a rerun my validation script

Thu, Mar 26 at 15:39

Lance Kuo-Esser

LK

so can we stick a drug in your DDG calculator?

Shayne Skrtic

SS

I actually got moved to a new department at work so I've been slammed busy but I should have some time to revalidate and update the parameters this weekend

Lance Kuo-Esser

no problem

LK

hopefully they're paying you more

Thu, Mar 26 at 18:38

Shayne Skrtic

SS

Made one python script and they just started throwing money at me. Industry is so goofy

Lance Kuo-Esser

they were like oh my god shane sit on my face

martin wishes that was him

LK

but children

Shayne Skrtic

SS

Overlap of comp. Sci and pediatrics is historically a pretty bad look can't lie

I'm going to Hawaii so I'm set

Tue, Apr 14 at 13:37

Lance Kuo-Esser

LK

i built an AI model to predict whether a amino acid sequence has a zinc finger for my class, so it's either yes or no. my model is 10% accurate, 40% worse than random guessing for fuck sake

1 Reply

shit i forgot to do stratified splitting for testing training set

LK

false alarm

Tue, Apr 14 at 15:12

LK

i built an AI model to predict whether a amino acid sequence has a zinc finger for my class, so it's either yes or no. my model is 10% accurate, 40% worse than random guessing for fuck sake

Shayne Skrtic

SS

Result = Reverse(result)

Shayne Skrtic

SS

Instant 90%

Wed, Apr 15 at 10:33

Lance Kuo-Esser

arXiv

Drug Discovery SMILES-to-Pharmacokinetics Diffusion
Models with Deep Molecular Understanding
arxiv.org

LK ok how bad would it be if we used this guy to aggregate the data?

Shayne Skrtic

SS I like it. If we use that for as many features as possible then do PCA we can see if the features actually help or not and if they don't we can just go back to plan A.

Thu, Apr 16 at 14:50

Lance Kuo-Esser

SwissADME
swissadme.ch



LK

Monday 14:58

Lance Kuo-Esser

LK alright when do we want to hop on a call in the next week or two?

Yesterday 10:35

Sure. I'm free this week. Next week I'm busy moving but then I'll be free the first week and a half in June

Lance Kuo-Esser

June 1 7PM?

LK EST

Sure

Lance Kuo-Esser

LK

anyone got unlimited zoom or are we gonna have to stoop down to teams

UHawaii uses the Google suite



Yesterday 12:51

Lance Kuo-Esser

LK

that is so much worse martin

Shayne Skrtic

SS

Real id rather hop on a Snapchat call than google meet

Lance Kuo-Esser

LK

emily has a fucking android and we have to FaceTime with that piece of shit

Tf. You are dating an android user?



Yesterday 15:06

Lance Kuo-Esser

LK

martin at least i'm dating someone

Yesterday 17:13

